PROJECT TITLE

HOSPITAL MANAGEMENT SYSTEM

A project report

Submitted in partial fulfilment of the requirements

for the award of the degree of

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Project Background

The health care industry requires an effective way of handling large amounts of data related to their patients, doctors, appointment schedules, billings, and hospital staff. Traditionally, all the hospitals had been recording the above data using the conventional way, which leads to data duplication, erroneous data, difficulty in accessing the data, and poor data organisation.

Due to advancements in technology, database management systems have become essential in handling the recording, organisation, and management of data in an effective manner. In order to address the shortcomings of the conventional method used in handling the data manually, Hospital Management Systems have been developed.

The study is aimed at developing a Hospital Management System using MySQL. The system involves numerous tables related to each other, including the Patients table, the Doctors table, the Appointment table, the Billings table, and the Staff table. The proposed system will effectively manage data storage and retrieval, appointment booking by patients for the doctor, billing information, and hospital staff.

Introduction of this system will significantly improve the performance of the hospitals in handling operations. Moreover, the introduction of the system will reduce instances of data duplication and erroneous data while improving decision-making efficiency.

Description of the Project

The Hospital Management System is a Database Management System (DBMS) that handles hospital information. The Hospital Management System is capable of carrying out various operations such as patients' information management, doctors' information management, appointments, billing, and employees' information management in a systematic manner.

The Hospital Management System comprises several tables related to each other; these include Patient Table, Doctors Table, Appointments Table, Billing Table, and Staff Table. Each table holds its own data.

Key Components of the System

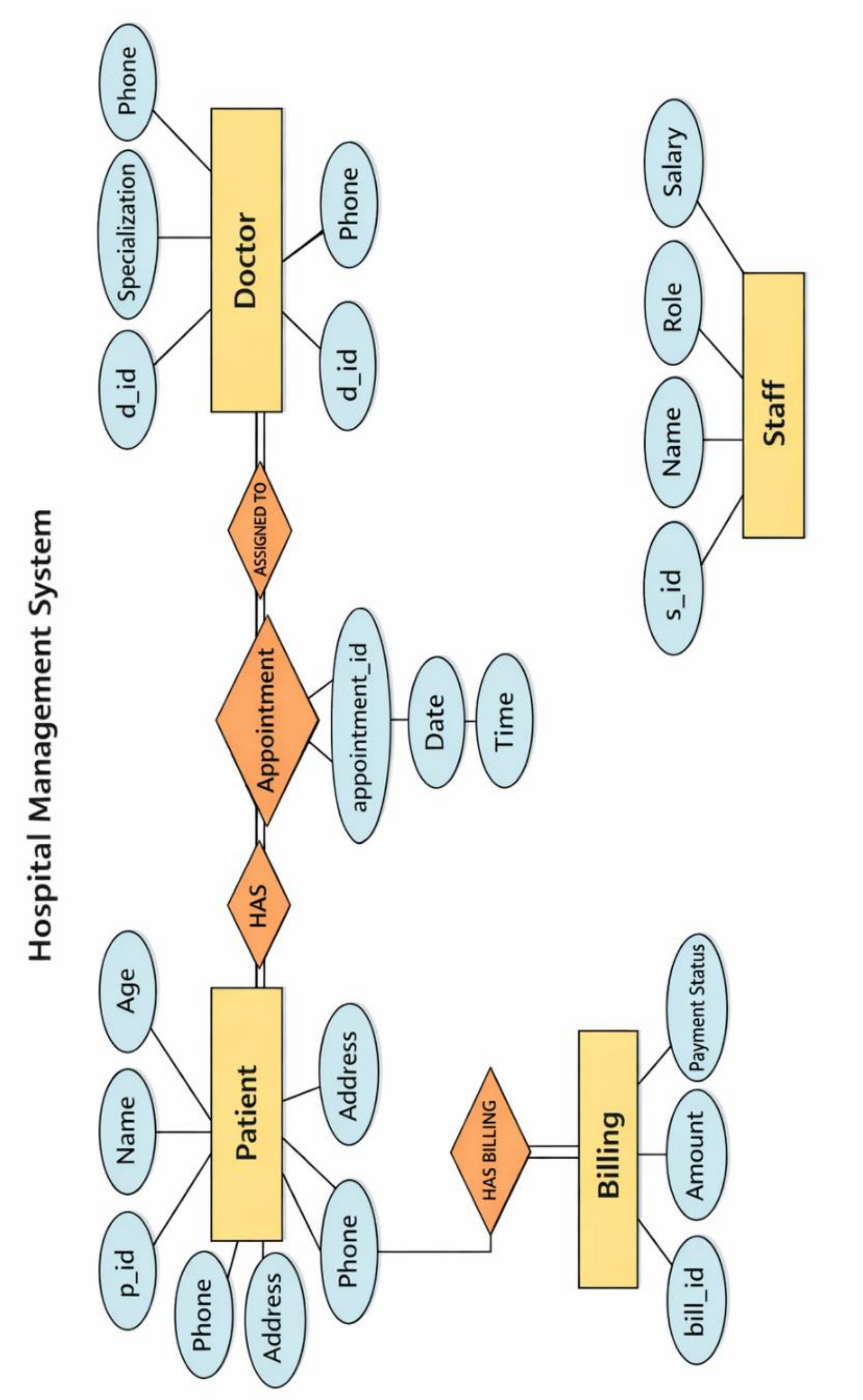
- **Patient Table**
  Stores personal details of patients such as name, age, gender, phone number, and address.
- **Doctor Table**
  Contains information about doctors including their name, specialization, and contact details.
- **Appointment Table**
  Manages scheduling of appointments and connects patients with doctors using foreign keys.
- **Billing Table**
  Maintains financial records including bill amount and payment status (Paid/Unpaid).
- **Staff Table**
  Stores details of hospital staff such as role, salary, and contact information.

The proposed system will offer effective storage and management of hospital data, and thus, it will be possible to retrieve data of patients and doctors easily. The system offers scheduling and monitoring of appointments. In addition, there are provisions in the database to handle proper billing and payment records. In addition, hospital staff data may be handled using the database.

Some of the operations done in the database include adding a record, updating a record, deleting a record, and selecting a record among others.

The introduction of some advanced features will improve the functionality of the database. Examples include JOIN operation, GROUP BY operation, creation of views, application of stored procedures, and implementation of triggers.On average, the proposed system will simplify the activities since it automates activities. It will save time and minimize paperwork. Besides, it increases the speed of hospital operations since one can retrieve information easily.

Entities - Relationships Diagram



Description of ER diagram

ER Diagram of the Hospital Management System

The ER Diagram of the Hospital Management System represents how the database is organized and how different entities within the Hospital Management System relate to each other.

There are five entities involved in this ER diagram, such as:

- **Patient:**
  The entity Patient contains information about the patient, including patient ID, name, age, gender, and other data. In this case, patient ID is set as the primary key.
- **Doctor:**
  The entity Doctor contains information about the physician, including doctor ID, name, specialty, and phone number. Here, doctor ID serves as the primary key.
- **Appointment:**
  The entity Appointment involves interaction between the patient and the doctor. The attributes in the entity are appointment ID, date, and time, meaning that there are two foreign keys in this particular situation.
- **Billing:**
  The entity Billing deals with billing information that belongs to the patients. In other words, its attributes include bill ID, total amount, and payment status, where the bill ID serves as the primary key.
- **Staff:**
  The Staff entity stores information about hospital employees, including their ID, name, role, salary, and contact details.
  It helps in managing and organizing staff-related data efficiently within the hospital system.

The relationships shown in the ER Diagram are as follows:

- There can be many Appointments for a Patient, but each Appointment is associated with only one Patient (One-to-Many).
- There can be many Appointments handled by a Doctor, but each Appointment is handled by only one Doctor (One-to-Many).
- There can be many Billing records for a Patient, but each Billing record belongs to only one Patient (One-to-Many).
- The Staff entity stores hospital employee details and functions independently without direct relationships with other entities.

Conversion of ER diagram into Tables and Description of Tables

The mapping of Entity-Relationship Diagram into relational tables is critical in database modeling. Through the ER diagram into relational tables conversion, each entity within the ERD is changed into a table and the associations among the entities are depicted through primary key and foreign key.

For the Hospital Management System, the key entities captured from the ER diagram include patient, doctor, appointment, billing, and staff. The entities are changed into distinct tables that will have their characteristics represented as the columns of the table. Each table will be assigned a unique identifier called primary key.

The entity patient is transformed into the Patient table that will contain attributes like p_id(primary key), name, age, gender, phone, and address. Likewise, the entity Doctor is mapped to the Doctor table which includes d_id(primary key), name, specialization, and phone. The staff entity will be transformed to Staff table including attributes such as s_id, name, salary, role, and phone.

The relations among the entities are taken care of during the transformation stage. For example, the Appointment entity, which is basically the relation between Patient and Doctor, will be transformed into the Appointment table consisting of fields like appointment_id (primary key), date, and time. There will also be foreign keys in the table to establish a connection with the Patient and Doctor tables, using fields like p_id and d_id.

In the same way, the Billing entity will become the Billing table with fields like bill_id (primary key), amount, and payment status. The relation between Patient and Billing will be established through the p_id field in the Billing table.

In this way, we get our ER diagram transformed into relational tables.

Description of Tables

The Hospital Management System database consists of five main tables: Patient, Doctor, Appointment, Billing, and Staff. Each table is designed to store specific information and is connected with other tables through primary and foreign keys.

The **Patient table** stores the personal and contact details of patients. It includes attributes such as p_id (primary key), name, age, gender, phone, and address. This table helps in maintaining all patient-related information in an organized manner.

Column Name	**Data Type**	**Description**
p_id	INT (PK)	Unique ID of the patient
name	VARCHAR(50)	Name of the patient
age	INT	Age of the patient
gender	VARCHAR(10)	Gender of the patient
phone	VARCHAR(15)	Contact number of the patient
address	VARCHAR(100)	Address of the patient

The **Doctor table** contains information about doctors working in the hospital. It includes attributes such as d_id (primary key), name, specialization, and phone. This table helps in identifying doctors and their areas of expertise.

Column Name	**Data Type**	**Description**
d_id	INT (PK)	Unique ID of the doctor
name	VARCHAR(50)	Name of the doctor
specialization	VARCHAR(50)	Doctor’s specialization
phone	VARCHAR(15)	Contact number of the doctor

The **Staff table** is used to store details of hospital employees other than doctors. It includes attributes such as s_id (primary key), name, role, salary, and phone. This table helps in managing staff-related information efficiently.

Column Name	**Data Type**	**Description**
s_id	INT (PK)	Unique ID of the staff
name	VARCHAR(50)	Name of the staff
role	VARCHAR(30)	Job role (Nurse, Receptionist)
salary	DECIMAL(10,2)	Salary of the staff
phone	VARCHAR(15)	Contact number of the staff

The **Appointment table** is used to manage the scheduling of appointments between patients and doctors. It includes attributes such as appointment_id (primary key), date, and time, along with p_id and d_id as foreign keys. These foreign keys establish relationships with the Patient and Doctor tables.

Column Name	**Data Type**	**Description**
appointment_id	INT (PK)	Unique ID of the appointment
p_id	INT (FK)	References Patient(p_id)
d_id	INT (FK)	References Doctor(d_id)
date	DATE	Date of the appointment
time	TIME	Time of the appointment

The **Billing table** is used to maintain financial records of patients. It includes attributes such as bill_id (primary key), amount, and payment_status, along with p_id as a foreign key. This table helps in tracking payments and billing details for each patient.

Column Name	**Data Type**	**Description**
bill_id	INT (PK)	Unique ID of the bill
p_id	INT (FK)	References Patient(p_id)
amount	DECIMAL(10,2)	Total bill amount
payment_status	VARCHAR(20)	Payment status (Paid/Unpaid)

Normalisation of tables up to 3-NF

The normalisation technique is used in designing databases for minimizing data redundancy and improve data integrity. The Hospital Management System follows normalising data up to Third Normal Form (3NF).

- First Normal Form (1NF)

The following characteristics must be satisfied for a table to be considered to be in 1NF:

All attributes should have only atomic values

All records should be unique

There shouldn't be any repeating groups or multi-valued attributes

For this project, all the tables like Patient, Doctor, Appointment, Billing, and Staff fall under 1NF since all columns have atomic values and all rows are uniquely identifiable with a primary key.

- Second Normal Form (2NF)

For a table to be considered in 2NF,

It should be already in 1NF

All non-primary key attributes should be completely dependent on the primary key

In this case, all tables have a primary key like p_id, d_id, etc., and all attributes completely rely on the primary key of a table. Thus, all tables are in 2NF.

- Third Normal Form (3NF)

Third Normal Form is satisfied when the following conditions are met:

It satisfies the requirements of 2NF

There are no transitive dependencies (attributes do not depend on any other attribute except the key)

In our case, all the non-key attributes depend on the primary key alone and not any other attribute. In the Patient table, the attributes such as name, age, and phone depend upon the primary key p_id only. This is true in the case of other tables where the attributes depend directly on the primary key. Thus, all the tables satisfy the criteria for Third Normal Form.

- Conclusion of Normalization

The Hospital Management System database is successfully normalized till Third Normal Form (3NF). It ensures:

No data redundancy

Data consistency

Efficient storage/retrieval

Creation of Data in the tables (at least 5 tables)

-Patient Table Schema and Structure

```
mysql> create database hospital;
Query OK, 1 row affected (0.04 sec)

mysql> use hospital
Database changed
mysql> create table patient ( p_id int primary key, name varchar(100) , age int, gender varchar(20), phone_num varchar(15), address varchar(100) );
Query OK, 0 rows affected (0.06 sec)

mysql> describe patient;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| p_id     | int          | NO   | PRI | NULL    |       |
| name     | varchar(100) | YES  |     | NULL    |       |
| age      | int          | YES  |     | NULL    |       |
| gender   | varchar(20)  | YES  |     | NULL    |       |
| phone_num | varchar(15) | YES  |     | NULL    |       |
| address  | varchar(100) | YES  |     | NULL    |       |
+----------+--------------+------+-----+---------+-------+
6 rows in set (0.03 sec)
```

Data insertion

```
mysql> INSERT INTO Patient (p_id, name, age, gender, phone_num, address) VALUES
    -> (1, 'Ravi', 25, 'Male', '9876543210', 'Vijayawada'),
    -> (2, 'Sita', 30, 'Female', '9123456780', 'Hyderabad'),
    -> (3, 'Arjun', 40, 'Male', '9012345678', 'Chennai'),
    -> (4, 'Ram', 50, 'Male', '4478365821', 'Delhi'),
    -> (5, 'Leena', 37, 'Female', '9946748267', 'Ongol');
ERROR 1062 (23000): Duplicate entry '1' for key 'patient.PRIMARY'
mysql> INSERT INTO Patient (p_id, name, age, gender, phone_num, address) VALUES
    -> (6, "Harshitha", 20, "female" , "9948913057", "vijayawada");\
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM PATIENT;
+------+-----------+------+--------+------------+------------+
| p_id | name      | age  | gender | phone_num  | address    |
+------+-----------+------+--------+------------+------------+
|    1 | Ravi      |   25 | Male   | 9876543210 | Vijayawada |
|    2 | Sita      |   30 | Female | 9123456780 | Hyderabad  |
|    3 | Arjun     |   40 | Male   | 9012345678 | Chennai    |
|    4 | Ram       |   50 | Male   | 4478365821 | Delhi      |
|    5 | Leena     |   37 | Female | 9946748267 | Ongol      |
|    6 | Harshitha |   20 | female | 9948913057 | vijayawada |
+------+-----------+------+--------+------------+------------+
6 rows in set (0.00 sec)
```

-Doctor Table Schema and Structure

```
mysql> create table doctor ;
ERROR 4028 (HY000): A table must have at least one visible column.
mysql> create table doctor ( d_id int primary key, name varchar (100), specialization varchar(100), phone varchar (15));
Query OK, 0 rows affected (0.02 sec)

mysql> decsribe doctor;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'd
ecsribe doctor' at line 1
mysql> describe doctor;
+----------------+--------------+------+-----+---------+-------+
| Field          | Type         | Null | Key | Default | Extra |
+----------------+--------------+------+-----+---------+-------+
| d_id           | int          | NO   | PRI | NULL    |       |
| name           | varchar(100) | YES  |     | NULL    |       |
| specialization | varchar(100) | YES  |     | NULL    |       |
| phone          | varchar(15)  | YES  |     | NULL    |       |
+----------------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

Data insertion

```
mysql>  INSERT INTO Doctor (d_id, name, specialization, phone) VALUES
    -> (106, 'Dr. Kumar', 'Dermatologist', '9011122233');
Query OK, 1 row affected (0.01 sec)

mysql> select * from doctor;\
+------+------------+----------------+------------+
| d_id | name       | specialization | phone      |
+------+------------+----------------+------------+
|  101 | Dr. Sharma | Cardiologist   | 9000000001 |
|  102 | Dr. Reddy  | Neurologist    | 9000000002 |
|  103 | Dr. Mehta  | Orthopedic     | 9000000003 |
|  104 | Dr. Roa    | Gynecologist   | 7947894638 |
|  105 | Dr. Swetha | Neurologist    | 8746375683 |
|  106 | Dr. Kumar  | Dermatologist  | 9011122233 |
+------+------------+----------------+------------+
6 rows in set (0.00 sec)
```

-Billing Table Schema and Structure

```
mysql> create table billing (bill_id int primary key, p_id int , amount decimal(10,2), payment_status varchar(20), foreign key (p_id) references patient(p_i
d) );
Query OK, 0 rows affected (0.06 sec)

mysql> describe billing;
+----------------+---------------+------+-----+---------+-------+
| Field          | Type          | Null | Key | Default | Extra |
+----------------+---------------+------+-----+---------+-------+
| bill_id        | int           | NO   | PRI | NULL    |       |
| p_id           | int           | YES  | MUL | NULL    |       |
| amount         | decimal(10,2) | YES  |     | NULL    |       |
| payment_status | varchar(20)   | YES  |     | NULL    |       |
+----------------+---------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

Data insertion

```
mysql> INSERT INTO Billing (bill_id, p_id, amount, payment_status) VALUES
    -> (5001, 1, 2000.00, 'Paid'),
    -> (5002, 2, 3500.00, 'Unpaid'),
    -> (5003, 3, 1500.00, 'Paid'),
    -> (5004, 4, 4200.00, 'Unpaid'),
    -> (5005, 5, 2800.00, 'Paid');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from billing;
+---------+------+---------+----------------+
| bill_id | p_id | amount  | payment_status |
+---------+------+---------+----------------+
|    5001 |    1 | 2000.00 | Paid           |
|    5002 |    2 | 3500.00 | Unpaid         |
|    5003 |    3 | 1500.00 | Paid           |
|    5004 |    4 | 4200.00 | Unpaid         |
|    5005 |    5 | 2800.00 | Paid           |
+---------+------+---------+----------------+
5 rows in set (0.00 sec)
```

-Appointment Table Schema and Structure

```
mysql> create table appointment (appointment_id int primary key, p_id int, d_id int , date Date, time Time ,foreign key (p_id) references patient(p_id) , fo
reign key (d_id) references doctor (d_id) );
Query OK, 0 rows affected (0.06 sec)

mysql> describe appointment;
+----------------+------+------+-----+---------+-------+
| Field          | Type | Null | Key | Default | Extra |
+----------------+------+------+-----+---------+-------+
| appointment_id | int  | NO   | PRI | NULL    |       |
| p_id           | int  | YES  | MUL | NULL    |       |
| d_id           | int  | YES  | MUL | NULL    |       |
| date           | date | YES  |     | NULL    |       |
| time           | time | YES  |     | NULL    |       |
+----------------+------+------+-----+---------+-------+
5 rows in set (0.01 sec)

mysql> |
```

Data insertion

```
mysql> INSERT INTO Appointment (appointment_id, p_id, d_id, date, time) VALUES
    -> (1008, 3, 106, '2026-04-27', '11:45:00');
Query OK, 1 row affected (0.01 sec)

mysql> select * from appointment;
+----------------+------+------+------------+----------+
| appointment_id | p_id | d_id | date       | time     |
+----------------+------+------+------------+----------+
|           1001 |    1 |  101 | 2026-04-20 | 10:30:00 |
|           1002 |    2 |  102 | 2026-04-21 | 11:00:00 |
|           1003 |    3 |  103 | 2026-04-22 | 12:00:00 |
|           1004 |    4 |  104 | 2026-04-23 | 09:45:00 |
|           1005 |    5 |  105 | 2026-04-24 | 01:15:00 |
|           1006 |    1 |  102 | 2026-04-25 | 02:00:00 |
|           1007 |    2 |  105 | 2026-04-26 | 03:30:00 |
|           1008 |    3 |  106 | 2026-04-27 | 11:45:00 |
+----------------+------+------+------------+----------+
8 rows in set (0.00 sec)
```

-Staff Table Schema and Structure

```
mysql> use hospital
Database changed
mysql> create table staff( staff_id int primary key, name varchar(100), role varchar(50), phone varchar(15), salary decimal(10,2));
Query OK, 0 rows affected (0.10 sec)

mysql> describe staff;
+----------+---------------+------+-----+---------+-------+
| Field    | Type          | Null | Key | Default | Extra |
+----------+---------------+------+-----+---------+-------+
| staff_id | int           | NO   | PRI | NULL    |       |
| name     | varchar(100)  | YES  |     | NULL    |       |
| role     | varchar(50)   | YES  |     | NULL    |       |
| phone    | varchar(15)   | YES  |     | NULL    |       |
| salary   | decimal(10,2) | YES  |     | NULL    |       |
+----------+---------------+------+-----+---------+-------+
5 rows in set (0.05 sec)
```

Data insertion

```
mysql> INSERT INTO Staff VALUES
    -> (201, 'Anita', 'Nurse', '9876500001', 25000.00),
    -> (202, 'Rahul', 'Receptionist', '9876500002', 20000.00),
    -> (203, 'Kiran', 'Ward Boy', '9876500003', 18000.00),
    -> (204, 'Sneha', 'Lab Technician', '9876500004', 30000.00),
    -> (205, 'Vikram', 'Pharmacist', '9876500005', 28000.00);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from staff;
+----------+--------+----------------+------------+----------+
| staff_id | name   | role           | phone      | salary   |
+----------+--------+----------------+------------+----------+
|      201 | Anita  | Nurse          | 9876500001 | 25000.00 |
|      202 | Rahul  | Receptionist   | 9876500002 | 20000.00 |
|      203 | Kiran  | Ward Boy       | 9876500003 | 18000.00 |
|      204 | Sneha  | Lab Technician | 9876500004 | 30000.00 |
|      205 | Vikram | Pharmacist     | 9876500005 | 28000.00 |
+----------+--------+----------------+------------+----------+
5 rows in set (0.00 sec)
```

SQL Queries Using Subqueries, Aggregate Functions, and Joins

- billing details with name

```
mysql> SELECT b.bill_id,
    ->        p.name AS patient_name,
    ->        b.amount,
    ->        b.payment_status
    -> FROM Billing b
    -> JOIN Patient p ON b.p_id = p.p_id;
+---------+--------------+---------+----------------+
| bill_id | patient_name | amount  | payment_status |
+---------+--------------+---------+----------------+
|    5001 | Ravi         | 2000.00 | Paid           |
|    5002 | Sita         | 3500.00 | Unpaid         |
|    5003 | Arjun        | 1500.00 | Paid           |
|    5004 | Ram          | 4200.00 | Unpaid         |
|    5005 | Leena        | 2800.00 | Paid           |
+---------+--------------+---------+----------------+
5 rows in set (0.00 sec)
```

- doctor wise patient list

```
mysql> SELECT d.name AS doctor_name,
    ->        p.name AS patient_name
    -> FROM Doctor d
    -> JOIN Appointment a ON d.d_id = a.d_id
    -> JOIN Patient p ON a.p_id = p.p_id
    -> ORDER BY d.name;
+-------------+--------------+
| doctor_name | patient_name |
+-------------+--------------+
| Dr. Mehta   | Arjun        |
| Dr. Reddy   | Sita         |
| Dr. Reddy   | Ravi         |
| Dr. Roa     | Ram          |
| Dr. Sharma  | Ravi         |
| Dr. Swetha  | Leena        |
| Dr. Swetha  | Sita         |
+-------------+--------------+
7 rows in set (0.01 sec)
```

- patient, doctor , appointment details

```
mysql> SELECT p.name AS Patient, d.name AS Doctor, a.date, a.time
    -> FROM Patient p
    -> JOIN Appointment a ON p.p_id = a.p_id
    -> JOIN Doctor d ON a.d_id = d.d_id;
+---------+------------+------------+----------+
| Patient | Doctor     | date       | time     |
+---------+------------+------------+----------+
| Ravi    | Dr. Sharma | 2026-04-20 | 10:30:00 |
| Ravi    | Dr. Reddy  | 2026-04-25 | 02:00:00 |
| Sita    | Dr. Reddy  | 2026-04-21 | 11:00:00 |
| Sita    | Dr. Swetha | 2026-04-26 | 03:30:00 |
| Arjun   | Dr. Mehta  | 2026-04-22 | 12:00:00 |
| Ram     | Dr. Roa    | 2026-04-23 | 09:45:00 |
| Leena   | Dr. Swetha | 2026-04-24 | 01:15:00 |
+---------+------------+------------+----------+
7 rows in set (0.00 sec)
```

- payment status update

```
mysql> UPDATE Billing
    -> SET payment_status = 'Paid'
    -> WHERE bill_id = 5002;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> select * from billing;
+---------+------+---------+----------------+
| bill_id | p_id | amount  | payment_status |
+---------+------+---------+----------------+
|    5001 |    1 | 2000.00 | Paid           |
|    5002 |    2 | 3500.00 | Paid           |
|    5003 |    3 | 1500.00 | Paid           |
|    5004 |    4 | 4200.00 | Unpaid         |
|    5005 |    5 | 2800.00 | Paid           |
+---------+------+---------+----------------+
5 rows in set (0.00 sec)
```

- number of patients based on gender

```
mysql> SELECT gender, COUNT(*) AS total
    -> FROM Patient
    -> GROUP BY gender;
+--------+-------+
| gender | total |
+--------+-------+
| Male   |     3 |
| Female |     2 |
+--------+-------+
2 rows in set (0.01 sec)
```

- number of appointments per doctor

```
mysql> SELECT d.name AS doctor_name,
    ->        COUNT(a.appointment_id) AS total_appointments
    -> FROM Doctor d
    -> LEFT JOIN Appointment a ON d.d_id = a.d_id
    -> GROUP BY d.d_id, d.name;
+-------------+--------------------+
| doctor_name | total_appointments |
+-------------+--------------------+
| Dr. Sharma  |                  1 |
| Dr. Reddy   |                  2 |
| Dr. Mehta   |                  1 |
| Dr. Roa     |                  1 |
| Dr. Swetha  |                  2 |
+-------------+--------------------+
5 rows in set (0.01 sec)
```

- total bill amount for each patient

```
mysql> SELECT p.name AS patient_name,
    ->        SUM(b.amount) AS total_amount
    -> FROM Billing b
    -> JOIN Patient p ON b.p_id = p.p_id
    -> GROUP BY p.p_id, p.name;
+--------------+--------------+
| patient_name | total_amount |
+--------------+--------------+
| Ravi         |      2000.00 |
| Sita         |      3500.00 |
| Arjun        |      1500.00 |
| Ram          |      4200.00 |
| Leena        |      2800.00 |
+--------------+--------------+
5 rows in set (0.00 sec)
```

- number of doctors in each specialization

```
mysql> SELECT specialization,
    ->        COUNT(*) AS total_doctors
    -> FROM Doctor
    -> GROUP BY specialization;
+----------------+---------------+
| specialization | total_doctors |
+----------------+---------------+
| Cardiologist   |             1 |
| Neurologist    |             2 |
| Orthopedic     |             1 |
| Gynecologist   |             1 |
+----------------+---------------+
4 rows in set (0.00 sec)
```

- counting staff by role

```
mysql> SELECT role, COUNT(*)
    -> FROM Staff
    -> GROUP BY role;
+----------------+----------+
| role           | COUNT(*) |
+----------------+----------+
| Nurse          |        1 |
| Receptionist   |        1 |
| Ward Boy       |        1 |
| Lab Technician |        1 |
| Pharmacist     |        1 |
+----------------+----------+
5 rows in set (0.00 sec)
```

- total salary for each role

```
mysql> SELECT role, SUM(salary) AS total_salary
    -> FROM Staff
    -> GROUP BY role;
+----------------+--------------+
| role           | total_salary |
+----------------+--------------+
| Nurse          |     25000.00 |
| Receptionist   |     20000.00 |
| Ward Boy       |     18000.00 |
| Lab Technician |     30000.00 |
| Pharmacist     |     28000.00 |
+----------------+--------------+
5 rows in set (0.00 sec)
```

- Join Staff with Doctor To show nurses along with doctors

```
mysql> SELECT s.name AS staff_name,
    ->        s.role,
    ->        d.name AS doctor_name
    -> FROM Staff s
    -> JOIN Doctor d
    -> WHERE s.role = 'Nurse';
+------------+-------+-------------+
| staff_name | role  | doctor_name |
+------------+-------+-------------+
| Anita      | Nurse | Dr. Sharma  |
| Anita      | Nurse | Dr. Reddy   |
| Anita      | Nurse | Dr. Mehta   |
| Anita      | Nurse | Dr. Roa     |
| Anita      | Nurse | Dr. Swetha  |
+------------+-------+-------------+
5 rows in set (0.01 sec)
```

Creation of Views Using the Defined Tables

view of appointments:

```
mysql> create view Appointments as select a.appointment_id , p.name as Patient_name , d.name as Doctor_name, a.date , a.time from appointment a join patient
 p on a.p_id = p.p_id join doctor d on a.d_id = d.d_id;
Query OK, 0 rows affected (0.03 sec)

mysql> select * from appointmens;
ERROR 1146 (42S02): Table 'hospital.appointmens' doesn't exist
mysql> select * from appointments;
+----------------+--------------+-------------+------------+----------+
| appointment_id | Patient_name | Doctor_name | date       | time     |
+----------------+--------------+-------------+------------+----------+
|           1001 | Ravi         | Dr. Sharma  | 2026-04-20 | 10:30:00 |
|           1002 | Sita         | Dr. Reddy   | 2026-04-21 | 11:00:00 |
|           1003 | Arjun        | Dr. Mehta   | 2026-04-22 | 12:00:00 |
|           1004 | Ram          | Dr. Roa     | 2026-04-23 | 09:45:00 |
|           1005 | Leena        | Dr. Swetha  | 2026-04-24 | 01:15:00 |
|           1006 | Ravi         | Dr. Reddy   | 2026-04-25 | 02:00:00 |
|           1007 | Sita         | Dr. Swetha  | 2026-04-26 | 03:30:00 |
|           1008 | Arjun        | Dr. Kumar   | 2026-04-27 | 11:45:00 |
+----------------+--------------+-------------+------------+----------+
8 rows in set (0.01 sec)
```

billing view

```
mysql> create view Billing_view as select b.bill_id , p.name as patient_name , b.amount, b.payment_status from billing b join patient p on b.p_id = p.p_id;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from billing_view;
+---------+--------------+---------+----------------+
| bill_id | patient_name | amount  | payment_status |
+---------+--------------+---------+----------------+
|    5001 | Ravi         | 2000.00 | Paid           |
|    5002 | Sita         | 3500.00 | Paid           |
|    5003 | Arjun        | 1500.00 | Paid           |
|    5004 | Ram          | 4200.00 | Unpaid         |
|    5005 | Leena        | 2800.00 | Paid           |
+---------+--------------+---------+----------------+
5 rows in set (0.00 sec)
```

view of number of appointments each doctor has

```
mysql> CREATE VIEW Doctor_Appointment_Count AS
    -> SELECT d.d_id,
    ->        d.name AS doctor_name,
    ->        COUNT(a.appointment_id) AS total_appointments
    -> FROM Doctor d
    -> LEFT JOIN Appointment a ON d.d_id = a.d_id
    -> GROUP BY d.d_id, d.name;
Query OK, 0 rows affected (0.02 sec)

mysql> select * from doctor_appointment_count;
+------+-------------+--------------------+
| d_id | doctor_name | total_appointments |
+------+-------------+--------------------+
|  101 | Dr. Sharma  |                  1 |
|  102 | Dr. Reddy   |                  2 |
|  103 | Dr. Mehta   |                  1 |
|  104 | Dr. Roa     |                  1 |
|  105 | Dr. Swetha  |                  2 |
|  106 | Dr. Kumar   |                  1 |
+------+-------------+--------------------+
6 rows in set (0.01 sec)
```